Independent Real-Time Verification of Manual Valve Position as an Additional Protection Layer Against Misrouting During Product Transfers: A Monte Carlo Barrier-Effectiveness Assessment

Leon Sandler

Independent Researcher, Northbrook, Illinois, USA

ORCID: https://orcid.org/0009-0007-4584-808X

Corresponding author: sandler.leon@gmail.com

Highlights

- Independent position sensing cuts modeled undetected misrouting 14-fold
- Over assumed priors the reduction factor is 9.7 (5th–95th percentile 4.4–28)
- Operator response to alarms is the weakest link: 90% response gives 5.6-fold
- An 8% alarm tolerance stays near-optimal once alarm fatigue is included
- Exact binomial acceptance criteria are derived for a pilot of 1000 valves

Abstract

Manual valve line-ups before product transfers rely on rounds, radio confirmation and checklists, none of which observes valve position independently. We assess an additional protection layer: temporary, reassignable clamp-on sensors that measure the actual position of existing manual valves, identify the operator and report to a plant historian. A Monte Carlo model of the line-up covers skipped, wrong-valve, incomplete and imprecise actions, out-of-sequence steps, unauthorized movement, detection by existing checks, and sensor bias, noise, drift, slip, wrong assignment, availability, bypass and operator complacency; it is verified against closed-form

results and an automated test suite. With assumed inputs for a 14-valve route, the probability of at least one undetected error per transfer fell from 4.03% to 0.29% with direct stem sensing and NFC identity (14-fold), against 0.45% for handwheel turn counting, which also gave four times more nuisance alarms. A probabilistic sensitivity analysis over the assumed priors gave a median risk-reduction factor of 9.7 (5th–95th percentile 4.4–28), above 5 in 88% of draws; the miss rate of existing barriers, calibration bias and device availability were the main drivers. The factor fell to 5.6 if operators acted on only 90% of true alarms, and an alarm tolerance near 8% of stroke remained best once alarm fatigue was included. Separate tolerances for throttled valves, end-stop re-zeroing and coverage of neighboring valves were quantified, and exact binomial acceptance criteria for a pilot were derived. The inputs are assumed, not measured: the results are specification targets, not credited risk reduction.

Keywords: Manual valve verification; line-up error; human reliability; protection layer; Monte Carlo simulation; probabilistic sensitivity analysis; wireless position sensing

1. Introduction

Product transfers between tanks, plants and terminals depend on a correct valve line-up. Where valves are manual, the line-up is established by field operators who open and close a defined set of valves in sequence, and it is confirmed by inspection rounds, radio communication between control rooms and checklists. This arrangement is well established, but it observes valve position only indirectly, through the report of the person who handled the valve. Human reliability analysis treats omission of a step, selection of the wrong item and incomplete execution as distinct and persistent error modes [1–4], and quantification methods such as HEART assign them generic error probabilities that are modified by conditions of the task [5]. Analyses of process accidents repeatedly identify human and organizational factors [6], and

collections of process-plant case histories include accidents in which the position of a manual valve was a factor [7]. In a transfer line-up each error mode can lead to a closed loop, contamination of another tank or delivery of the wrong product to a receiving site.

Process safety practice already places protection layers between such errors and their consequences, including procedures, second-person checks, locked valves and instrumented position indication on automated valves [8,9]. For manual valves in remote tank areas, continuous independent position indication is often not installed, because permanent limit switches or position transmitters require wiring, hazardous-area installation effort and capital cost per valve. Retrofit wireless position monitoring of valves has been demonstrated [10], and industrial demand reflects the same gap: operators want the position, identity and time of every manual valve action visible in the plant historian without replacing or automating the valve, as illustrated by a recent open-innovation call for such a solution [11].

Quantitative evidence for such a layer is scarce, and a design-stage assessment has two obvious weaknesses. It rests on assumed human-error rates, and it tends to assume that the people involved react to every alarm. The first version of this study (retained unchanged in the repository) addressed the architecture and the main trade-offs under one set of assumed inputs. This paper extends it in five ways that bear on those weaknesses: (i) a probabilistic sensitivity analysis that propagates the uncertainty of the assumed inputs to the risk-reduction factor; (ii) a model of the operator's response to alarms, including alarm fatigue [12]; (iii) common-cause variation of the error rates; (iv) separate alarm tolerances for throttled valves, end-stop re-zeroing and coverage of neighboring valves, three design questions left open before; and (v) exact binomial acceptance criteria and sample sizes for a pilot that would replace the assumed inputs

with data. Each added mechanism reduces to the first version when it is switched off, and this is tested (Section 4.9).

The study makes three contributions. First, it describes a sensing-layer architecture for temporary, reassignable, non-invasive verification of manual valve position that combines position, operator identity, time and procedure step. Second, it presents a Monte Carlo framework that quantifies how such a layer changes the probability that an error goes undetected at pump start or during the transfer, with the false-alarm cost, and how robust that change is to the assumed inputs. Third, it derives specification guidance and pilot acceptance criteria from the simulated trade-offs. It is a design-stage assessment based on assumed inputs and reports no field data.

2. Problem definition

2.1 The line-up task

A representative task is an interplant transfer of a gasoline blend between a refinery and a logistics terminal. Before pumping, operators at both sites manually set a defined list of valves to required positions, mostly fully open or fully closed, with occasional intermediate settings. The route has V valves; this work examines routes of 6, 14 and 24 valves, and uses the 14-valve route (eight to be opened, four closed, two throttled) as the reference.

2.2 Error modes and existing barriers

Six error classes are considered: (i) a required valve action is skipped, including a false report that it was done; (ii) the wrong valve is operated and the intended valve is left unchanged; (iii) the movement stops short; (iv) an intermediate valve is set imprecisely; (v) steps are performed out of sequence; and (vi) a valve is moved without authorization after validation. Existing

barriers (inspection rounds, radio confirmation, paper checklist) detect each class with some probability, but none provides independent, time-stamped observation of the physical position.

2.3 Requirements for an additional layer

The additional layer should (a) measure actual position, including intermediate openings, independently of operator confirmation; (b) record movements and deviations with time stamps; (c) work on existing manual valves without modification or process shutdown; (d) be installable for a few days and reassigned unambiguously to another valve, with an assignment history; (e) suit classified areas; (f) tolerate wrench operation; (g) use existing industrial wireless infrastructure and feed a plant historian; and (h) alarm on loss of signal, battery or communication while buffering data locally. These requirements follow the problem statement of an industrial open-innovation call for such a solution [11].

3. Sensing-layer architecture

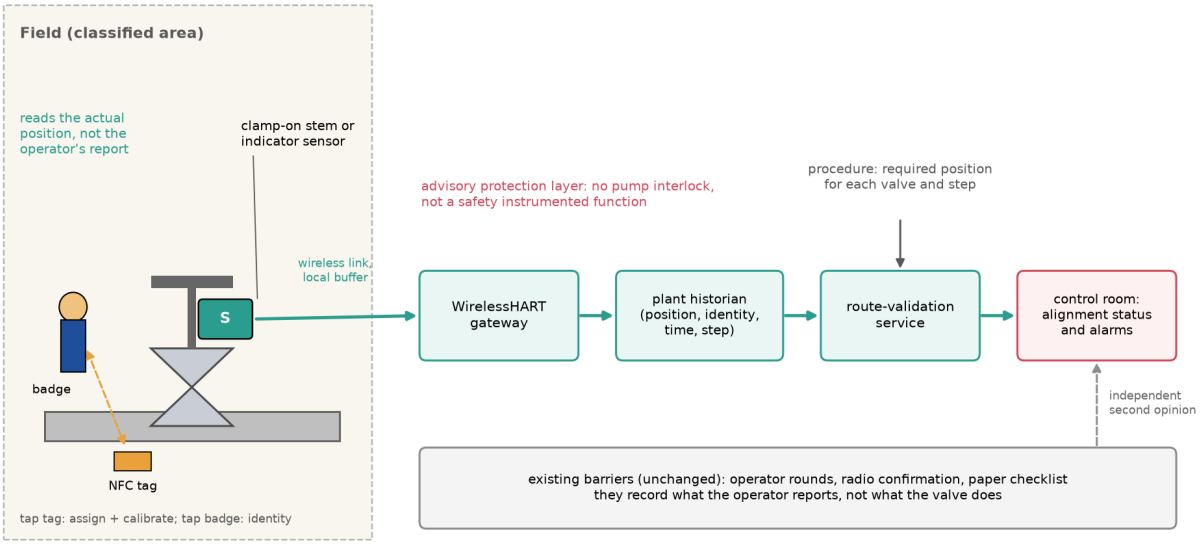


Fig. 1. Architecture of the sensing layer: a clamp-on sensor on the stem or indicator of an existing manual valve, an NFC tag for identity and assignment, a wireless link to the gateway

and historian, a route-validation service that compares required and measured positions, and the existing barriers, which are unchanged.

3.1 Measurement principle

Two measurement concepts are compared. In the handwheel turn-count collar, a clamp-on device on the handwheel counts rotations and converts turns to percent of a calibrated stroke. It is simple to mount on many valves, but position is inferred: gear-operator lash, slippage and lost counts accumulate error. In direct stem or indicator sensing, a clamp-on kit reads the angle of the stem or position indicator (rotary) or the stem travel (rising stem), quantities that move with the closure element. Direct sensing also needs calibration but is not exposed to handwheel slip. The device is mounted outside the tool path and shrouded, so operation with a wrench does not load the sensor.

3.2 Identity, assignment and event logic

A passive near-field communication (NFC) tag carrying the valve identity is fixed to each valve. Tapping the device on the tag assigns it to the valve; a guided two-point calibration (fully closed and fully open) follows; and a backend ledger stores assignments over time. Tapping the operator badge before touching a valve links each movement to a person. A mismatch between the tapped valve and the valve required by the current procedure step can alarm before the wrong valve is moved, which converts some detection into prevention.

3.3 Communication and integration

Each device reports through a certified WirelessHART link [13] to the plant gateway, and the gateway forwards data to the plant historian through standard industrial protocols. Movements are reported on event, with a periodic heartbeat. The device stores events locally and forwards

them when communication returns. A route-validation service compares required and measured positions and displays alignment status before pumping begins.

3.4 Role as a protection layer

The layer is advisory. It does not interlock pumps and is not claimed as a safety instrumented function under the functional-safety standards for the process sector [14]. Its value is an additional, independent detection opportunity for human errors. Hazardous-area use requires an appropriate protection concept such as intrinsic safety [15], which also limits the energy available to the electronics and motivates a low-power design.

4. Monte Carlo model

4.1 Overview

The model simulates N independent transfers. For each transfer, the actual position a_v of every route valve v is generated from the required position r_v and the error model of Section 4.2; existing barriers then detect some defects (Section 4.3); and the sensing layer measures positions and raises alarms (Section 4.4). A defect is defined as $|a_v - r_v| > 0.10$ of stroke. The Monte Carlo method follows its standard formulation [16], and probabilities are reported with Wilson 95% intervals [17,18].

4.2 Operator error model

Per valve action, the probabilities of skipping, operating the wrong valve and stopping short are 0.002, 0.0015 and 0.006. These are order-of-magnitude values in the range used in human reliability practice for procedural tasks [1,2], and they are assumed priors, not measurements. A skipped or wrong-valve action leaves the intended valve at its random initial state. An incomplete movement completes a fraction uniformly distributed between 0.60 and 0.97 of the required travel. Intermediate valves are set with a standard deviation of 0.04 of stroke. Out-of-

sequence steps occur with probability 0.004 per transfer. Unauthorized or accidental movement occurs at 5×10^{-5} per valve-hour over a lognormal transfer duration with median 8 h; its size is uniform between 0.10 and 1.0 of stroke.

4.3 Existing barriers

Existing checks detect a skipped, wrong-valve, incomplete, imprecise, out-of-sequence and post-validation defect with probabilities 0.80, 0.80, 0.60, 0.50, 0.10 and 0.20 respectively. Each probability is perturbed per transfer on the logit scale with standard deviation 0.5 to represent uncertainty. When a sensing layer is present, operator complacency reduces the effectiveness of the existing checks by a relative factor, taken as 0.20 in the base case, because operators may rely on the system.

4.4 Sensing layer

The measured position of valve v is $m_v=a_v+b+e+d\sqrt{k}+s$, where b is the calibration bias (standard deviation σ_c), e is noise (σ_n), d is drift per square root of operation for k operations since calibration, and s is a slip event that occurs with probability $1-(1-p_s)^k$ and has magnitude uniform between 0.05 and 0.30 of stroke with random sign. The device alarms if $|m_v-r_v|>\tau$, with $\tau=0.08$ in the base case. With probability p_a the device is linked to the wrong valve, so that it measures another route valve. A device is unavailable at validation with probability 0.006 (communication 0.003, battery 0.001, hardware 0.002); if any route device is unavailable the validation is blocked, and with probability 0.20 the operator proceeds without sensor validation (bypass). Table 1 lists the architecture parameters.

Parameter	Existing checks	Turn-count collar	Direct sensing	Direct + NFC
Noise σ_n (stroke)	–	0.020	0.015	0.015
Calibration bias σ_c (stroke)	–	0.030	0.020	0.020
Drift d (stroke per √operation)	–	0.0040	0.0005	0.0005
Slip probability p_s per operation	–	0.0040	0.0003	0.0003
Wrong-device assignment p_a	–	0.005	0.005	0.0002

Parameter	Existing checks	Turn-count collar	Direct sensing	Direct + NFC
NFC identity and tap-before-touch	no	no	no	yes

Table 1. Sensing architectures compared (assumed parameters).

4.5 Outputs and closed-form verification

The primary output is the probability per transfer that at least one defect, out-of-sequence step or post-validation movement escapes all barriers. Secondary outputs are false alarms (an alarm on a valve that is not defective), the share of true defects missed by the sensors, the probability that validation is blocked, and detection within 60 s of a post-validation movement. The simulation was checked against three closed-form results. For ideal Gaussian measurement with standard deviation σ, the false-alarm probability is $2[1-\Phi(\tau/\sigma)]$. For a defect of offset e, the miss probability is $\Phi((\tau-e)/\sigma)-\Phi((-\tau-e)/\sigma)$. For independent binary-valve defects detected by existing barriers with probabilities d_i, the residual risk is $1-\prod(1-q_i(1-d_i))$. Simulated and closed-form values were 0.770% and 0.766%, 0.286% and 0.257%, and 1.85% and 1.84%, respectively.

The base case uses 300,000 transfers per architecture and route, 120,000 per sweep point and fixed random seeds; it follows the structure of an open simulation framework previously published by the author [19].

4.6 Extensions of the model

Five mechanisms were added to the first version; each can be switched off. (i) Common cause: the three operator error rates of a transfer are multiplied by a lognormal factor with mean one and log-standard deviation σ_{cc}, which represents fatigue, poor lighting or a mislabeled valve group acting on all the valves of one transfer. (ii) Operator response: an alarm on a truly defective valve is acted on with probability $\rho=\rho_0\, exp(-\kappa F)$, where F is the number of nuisance alarms (alarms on valves that are not defective) in the same validation, ρ_0 is the response to an isolated alarm and κ is a fatigue coefficient; this is a parametric description of alarm fatigue [12],

not a fitted model, and it applies to all alarm-driven detections. (iii) Separate alarm tolerance for intermediate (throttled) valves. (iv) End-stop re-zeroing: when a binary valve reaches its end stop the sensor offset is re-zeroed, so that for those valves the operations since the last zero are at most one. (v) Collateral movement: a wrong-valve action moves a neighboring valve that carries no route device with probability 0.5; the neighbor carries a device with probability equal to the coverage c; a covered movement is detected unless the validation was bypassed and is then acted on with the response probability ρ, and an uncovered movement is detected only by the existing barriers (detection probability 0.50 for such events).

4.7 Probabilistic sensitivity analysis

The first version varied one input at a time. Here the 13 uncertain inputs of Table 3 are varied together. Each draw samples the inputs from the distributions of the table (lognormal multipliers whose 90% interval is the stated factor, and uniform distributions for the three policy and behavior inputs), simulates 40,000 transfers for each of the existing-checks, direct-sensing and direct-plus-NFC cases with the same seed, and records the residual risks and the risk-reduction factor $RRF = P_{\text{existing}}/P_{\text{full}}$. 200 draws were made. The percentiles of the draws describe the effect of the assumed uncertainty, not of the sampling (the sampling error of a proportion near 0.9 from 200 draws is about 0.02). Partial rank correlation coefficients (PRCC) between the inputs and the logarithm of the output rank the drivers [20]. The prior distributions are assumed; they are wider than a typical expert interval on purpose, because the model has no plant data to narrow them.

Input	Base value	Distribution
Operator error rates (skip, wrong, incomplete)	× 1	lognormal, 90% factor 4
Existing-barrier miss probability	× 1	lognormal, 90% factor 3
Out-of-sequence rate	0.004	lognormal, 90% factor 3
Unauthorized-movement rate	5e−5 per valve-hour	lognormal, 90% factor 3
Sensor noise	× 1	lognormal, 90% factor 2
Sensor calibration bias	× 1	lognormal, 90% factor 2
Sensor drift	× 1	lognormal, 90% factor 3
Slip rate	× 1	lognormal, 90% factor 3

Input	Base value	Distribution
Wrong-device assignment	× 1	lognormal, 90% factor 3
Device unavailability (communication, battery, hardware)	× 1	lognormal, 90% factor 4
Bypass probability when blocked	0.20	uniform 0–0.5
Operator complacency	0.20	uniform 0–0.6
Time-stamp sequence detection	0.95	uniform 0.7–1.0

Table 3. Input distributions of the probabilistic sensitivity analysis (assumed). A lognormal multiplier with 90% factor f has 5th and 95th percentiles 1/f and f.

4.8 Pilot sizing

The acceptance criteria of a pilot are exact binomial tests. For nuisance alarms, a pilot validates N valves and accepts the device if no more than c nuisance alarms occur, where c is the largest count whose probability under the unacceptable rate (3% per valve, twice the modeled rate) is at most 5% (consumer risk); the power is the probability of acceptance at the modeled 1.4% per valve. For missed defects, a number of defects injected on purpose (valves set deliberately off by more than the defect threshold) is detected or not, and the test is built the same way for an unacceptable miss rate of 15%, with the power evaluated at the modeled 6%. The accuracy of a device is described by the width of the chi-square interval for the standard deviation of repeated readings. The unacceptable rates are illustrative thresholds chosen for this study, not criteria set by any operator.

4.9 Verification and reproducibility

The extended engine imports the first version unchanged (its file hash is checked) and, with all new mechanisms off, reproduces its event counts exactly for the same seeds. A suite of 12 automated checks covers this equivalence, the closed-form cases of Section 4.5, the direction of every added mechanism (response, tolerance, per-type tolerance, re-zeroing, coverage), the exact binomial calculations of the pilot, the probabilistic sensitivity analysis and the reproduction of

the packaged results of the first version. One command regenerates every number and figure; the code, results and manuscript sources are archived with DOIs (Data availability).

5. Results

Sections 5.1 to 5.6 are the base-case results of the first version of the study (single set of assumed inputs); Sections 5.7 to 5.13 are the extended studies. All probabilities are per transfer unless stated, for the 14-valve route unless stated.

5.1 Residual risk by architecture

For the 14-valve route (Table 2, Fig. 2), existing checks alone leave a probability of 4.03% per transfer that an error escapes. A handwheel turn-count collar reduces it to 0.45%, direct stem sensing to 0.30% and direct sensing with NFC identity to 0.29%, a 14-fold reduction. The difference between direct sensing with and without NFC is 1.2 standard errors, so the NFC benefit for residual risk is not resolved here; it appears in prevention (Section 5.5). Absolute values depend on assumed error rates, but the ratios depend mainly on detection logic and are therefore more robust; Section 5.7 tests how robust.

Route	Existing checks	Turn-count collar	Direct sensing	Direct + NFC
6 valves	2.069%	0.215%	0.138%	0.136%
14 valves	4.027%	0.454%	0.302%	0.285%
24 valves	6.345%	0.786%	0.544%	0.510%

Table 2. Probability per transfer that an error escapes all barriers (300,000 transfers per cell).

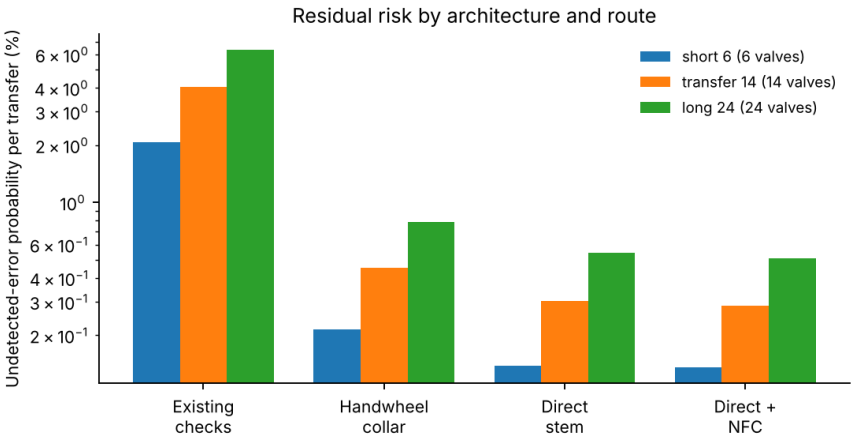


Fig. 2. Residual risk by architecture and route length.

Residual risk grows with route length for every architecture, from 0.14% for 6 valves to 0.51% for 24 valves with the full layer, because more valves give more opportunities for error and more exposure to device unavailability.

5.2 Alarm tolerance

Alarm tolerance controls the trade-off between nuisance alarms and missed defects (Fig. 3). With direct sensing at a 5% tolerance, false alarms reach 7.9% per valve, equivalent to about 1.1 alarms per 14-valve validation, which is unlikely to be acceptable under alarm-management practice, which limits alarm rates to what an operator can handle [21]. At 8%, false alarms drop to 1.4% per valve while 5.6% of true defects are missed, and these misses concentrate in defects only slightly larger than the 0.10 definition. At 10% the miss rate rises to 12.2%. For the turn collar at 8%, false alarms are 5.4% per valve, about four times higher than direct sensing. In the full-layer one-at-a-time analysis, tolerance moved the residual risk from 0.09% at half the base value to 1.96% at double, the largest effect of any parameter.

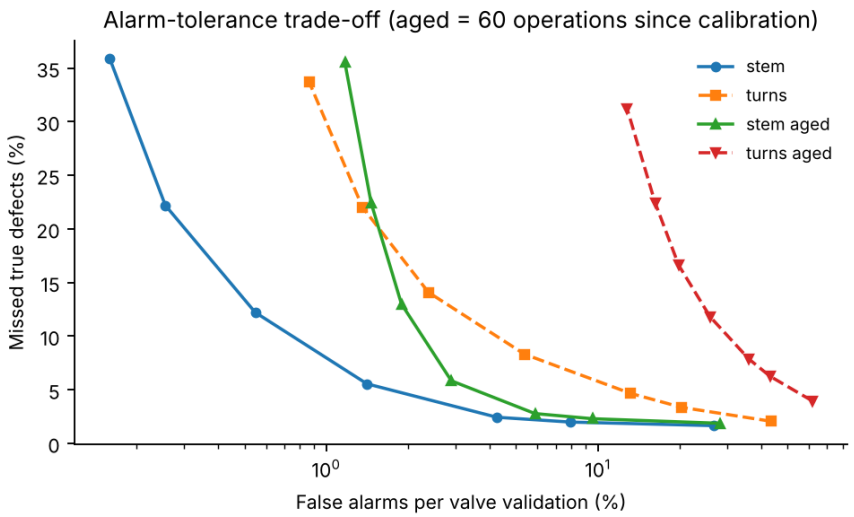


Fig. 3. False alarms versus missed defects as alarm tolerance varies; aged curves assume 60 operations since calibration.

5.3 Calibration age

Measurement quality degrades with operations since calibration (Fig. 4). For direct sensing, false alarms rise from 1.4% per valve at one operation to 2.1% at 30 and 8.7% at 300 operations. For the turn collar the corresponding values are 4.5%, 15.8% and 64.7%. A turn collar is therefore suitable only for short deployments calibrated at every assignment, whereas direct sensing tolerates about 100 operations between calibrations (3.9% false alarms per valve) and supports longer installations.

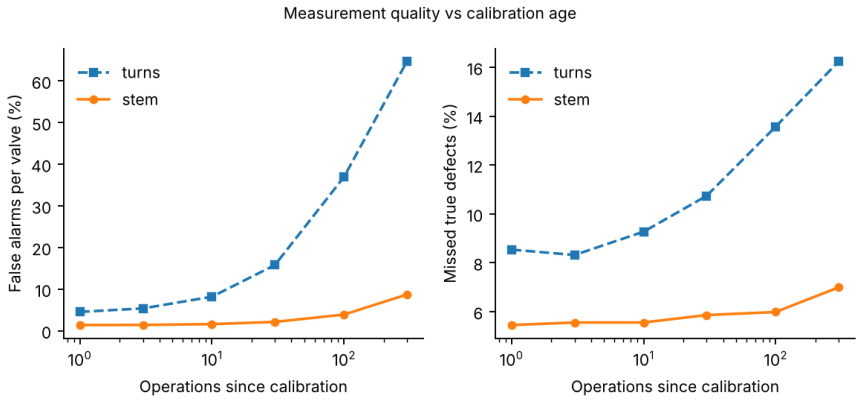


Fig. 4. False alarms and missed defects versus operations since calibration.

5.4 Availability and bypass

At the base device unavailability, validation is blocked in 8.0% of transfers on the 14-valve route and bypass occurs in 1.6% (Fig. 5). If bypass were prohibited, residual risk would fall to 0.21% regardless of communication quality in the range studied, while with 20% bypass it is 0.27% at the base communication-loss probability of 0.003 and 0.71% at 0.05. Blocking rises quickly with per-device communication loss, reaching 17% at 0.01. Availability is thus an operational constraint, and bypass policy determines whether it becomes a safety constraint.

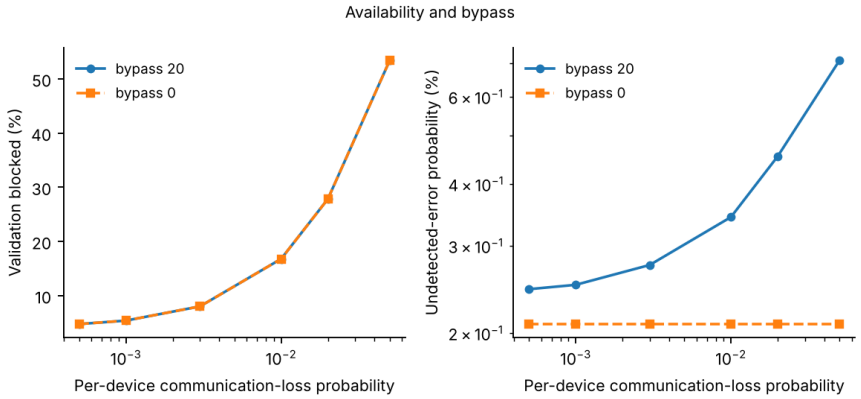


Fig. 5. Validation blocked and residual risk versus per-device communication loss, with and without bypass.

5.5 Operator complacency and identity checks

If operators relax existing checks because a sensing layer is present, residual risk rises from 0.229% with no complacency to 0.274% at 20% and 0.431% at 80% (Fig. 6), still far below 4.03% for existing checks alone, because the sensors observe independently. NFC tap-before-touch addresses a different problem: wrong-valve events, which move a valve that carries no device and are invisible to the sensing layer. At 90% tap compliance, wrong-valve events fall from 20.4 to 3.1 per 1000 transfers.

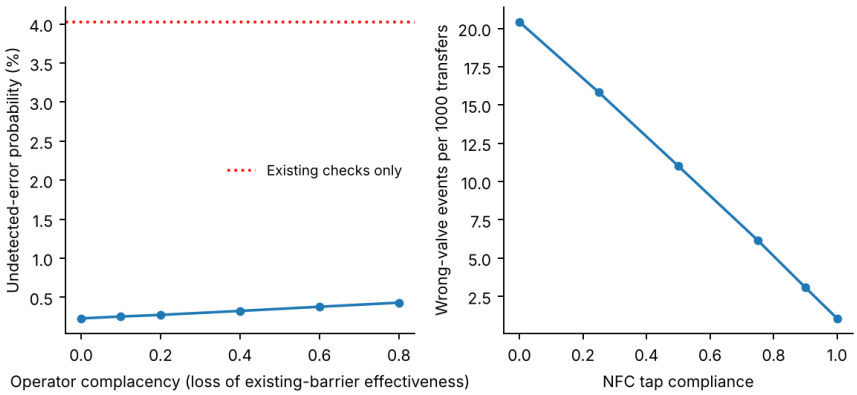


Fig. 6. Residual risk versus complacency (left) and wrong-valve events versus NFC tap compliance (right).

5.6 One-at-a-time sensitivity and device energy

In one-at-a-time sensitivity analysis over a factor 0.5 to 2 (Fig. 7), the ranking by effect on residual risk is alarm tolerance, calibration bias (0.18% to 0.48%), the miss rate of existing barriers (0.18% to 0.46%) and sensor noise, followed by bypass, availability and complacency. Operator error rates themselves have smaller effects once the sensing layer is present, because the layer detects defects regardless of how they arise. A design-level energy model (primary lithium cells, wake-on-motion, 16 s heartbeat) gives a median battery life of 4.8 years with a 5th

to 95th percentile range of 2.9 to 7.9 years (Supplementary Information); this must be confirmed by bench measurement.

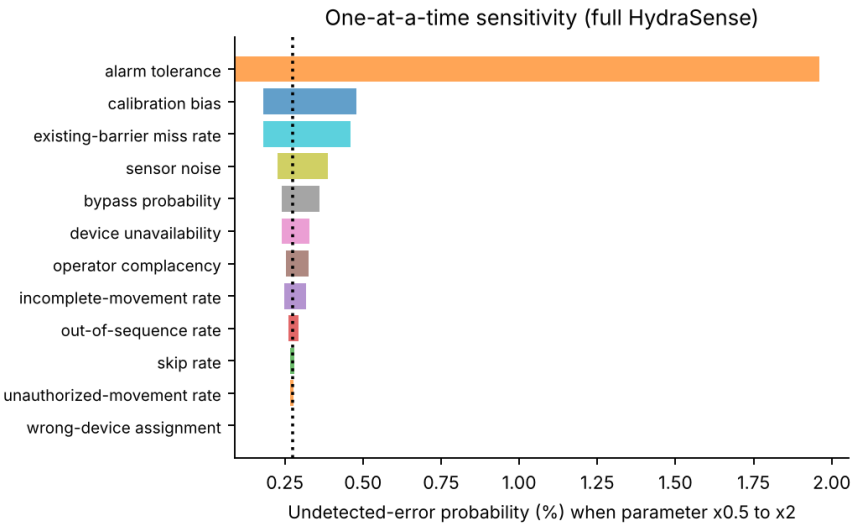


Fig. 7. One-at-a-time sensitivity of residual risk for the full layer (parameters varied by 0.5 and 2 times).

5.7 Uncertainty of the assumed inputs

Varying the 13 assumed inputs of Table 3 together (Fig. 8, Table 4) gives a median residual risk of 4.4% with existing checks alone (5th to 95th percentile 1.5% to 15%) and 0.42% with the full layer (0.17% to 1.17%). The risk-reduction factor has a median of 9.7 with a 5th to 95th percentile range of 4.4 to 28; it exceeds 2 in 100% of the draws, 5 in 88% and 10 in 46%. The nominal 14-fold reduction is exceeded in only 24% of the draws, and the reduction is better stated as roughly a factor of 10 with a plausible range from 4 to 28 than as a single value. The nuisance-alarm rate has a median of 1.5% per valve with an upper 95th percentile of 7.2%.

Quantity	5th percentile	Median	95th percentile
Existing checks only	1.50	4.42	15.15
Direct sensing	0.18	0.47	1.24
Direct sensing + NFC (full layer)	0.17	0.42	1.17
Nuisance alarms per valve (full layer)	0.83	1.46	7.23
Risk-reduction factor (existing / full)	4.4	9.7	28.4

Table 4. Probabilistic sensitivity analysis: percentiles of the residual risk per transfer (%) and of the risk-reduction factor over 200 draws of the assumed inputs (40,000 transfers per draw and case).

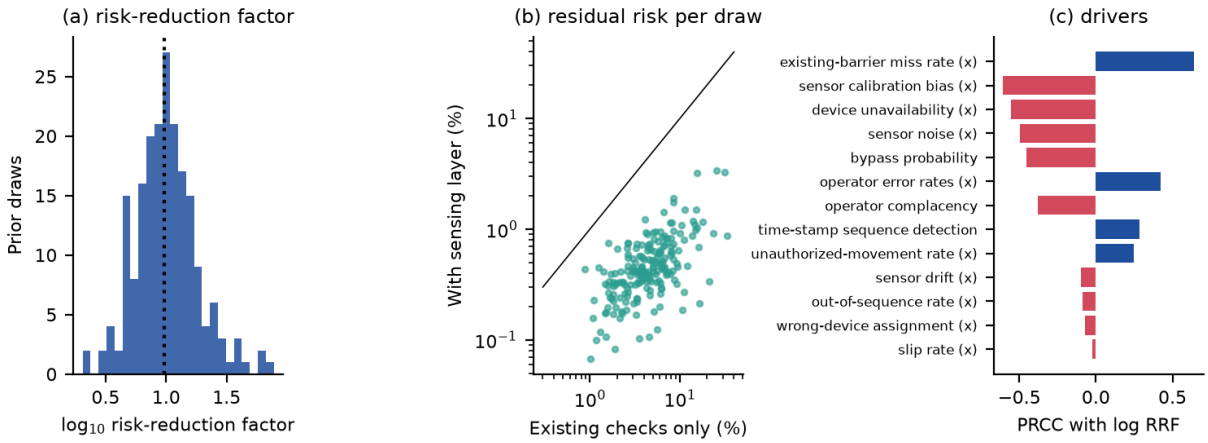


Fig. 8. Probabilistic sensitivity analysis over the assumed inputs. (a) Distribution of the risk-reduction factor; the dotted line is the median. (b) Residual risk with the sensing layer against existing checks only for each draw; the line is no reduction. (c) Partial rank correlation of each input with the logarithm of the risk-reduction factor.

The strongest rank correlations with the risk-reduction factor are the miss rate of the existing barriers (0.64), the sensor calibration bias (−0.61), device unavailability (−0.55), sensor noise (−0.50) and the bypass probability (−0.45). The sign has a simple reading: the more the existing barriers miss, or the more the operators err (0.42), the more the layer adds; the worse the sensing layer performs or the more often it is unavailable or bypassed, the less. Sensor calibration and availability, which a design can control, are therefore as important to the benefit as the unknown human error rates that the study cannot control.

5.8 Common-cause variation

Making the error rates of a transfer share a lognormal multiplier with mean one changes the per-transfer probability very little (4.06% to 3.92% for existing checks and 0.28% to 0.27% for the full layer as the log-standard deviation grows from 0 to 1.5, Fig. 9a), and the risk-reduction factor from 14.7 to 14.3. This is expected: a mean-preserving multiplier concentrates errors in fewer transfers but leaves the expected number almost unchanged, and the probability of at least one error is slightly reduced by the concentration. The independence assumption between valves therefore matters little for the headline probability, though it would matter for the probability of several simultaneous errors, which is not an output of this study.

5.9 Operator response and the choice of tolerance

If operators act on only part of the true alarms the benefit falls quickly (Fig. 9b gives the effect on the tolerance; the grid is in the Supplementary Information). With a response of 90% to an isolated alarm and no fatigue, the residual risk of the full layer rises from 0.27% to 0.72% and the risk-reduction factor falls from 14.8 to 5.6; with 80% response and a fatigue coefficient of 0.6 the factor is 2.9. A model without a response mechanism assumes a response of 100%; the extended model shows that the first result is the upper end of what a design can claim. The response also changes the optimum alarm tolerance. With perfect response, residual risk falls steadily as the tolerance tightens (0.09% at 4%), at the price of 2.1 nuisance alarms per validation, against 0.19 at 8%. With a response of 95% and fatigue coefficients of 0.3 and 0.6, the residual risk has its minimum at a tolerance of 8% of stroke for both coefficients (0.66% and 0.79%), because below it nuisance alarms teach the operator to ignore the true ones and above it defects go unflagged. The 8% tolerance used in the first version is therefore close to optimal

once alarm fatigue is included, but the shallow minimum and the dependence on the assumed response mean that the tolerance should be set by the pilot (Section 5.13).

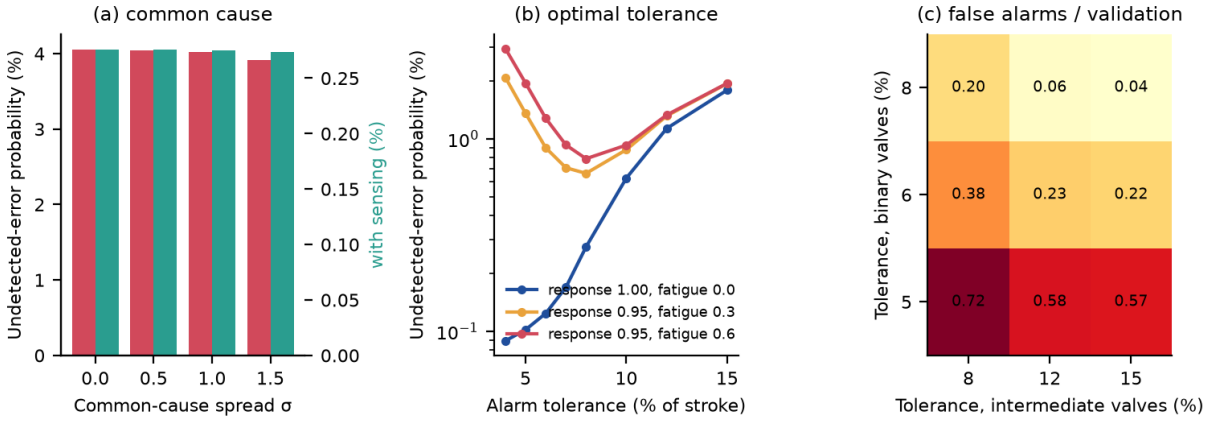


Fig. 9. (a) Common cause: undetected-error probability against the log-standard deviation of the shared multiplier. (b) Residual risk of the full layer against alarm tolerance for perfect and imperfect operator response. (c) Nuisance alarms per validation for direct sensing as the tolerances for binary and intermediate valves are varied separately.

5.10 Separate tolerance for throttled valves

With a common tolerance of 8%, the two throttled valves of the route produce 81% of the nuisance alarms: 8.1% per throttled valve against 0.3% per binary valve, because the operator's setting scatter (standard deviation 0.04 of stroke) is a large part of the 0.08 band. Widening the tolerance of the throttled valves alone to 12% cuts the nuisance alarms per validation from 0.20 to 0.06 (Fig. 9c), but it also raises the share of defects that are missed from 5.6% to 20.4% and the residual risk of direct sensing from 0.31% to 1.08%, because the defect threshold (0.10 of stroke) lies inside the wider band. The trade-off follows from the definition of a defect: if the process tolerates a wider band on a throttled valve, the defect threshold and the alarm band should be widened together; if it does not, the nuisance alarms are the cost of detecting a real defect and the design should reduce the setting scatter (a graduated scale, a detent) instead.

5.11 End-stop re-zeroing

Re-zeroing a binary valve each time it reaches an end stop removes most of the calibration-age effect for those valves (Fig. 10a): at 100 operations since calibration the nuisance alarms are 1.7% per valve with re-zeroing against 3.9% without, and at 300 operations 2.3% against 8.7%. The throttled valves cannot be re-zeroed at an end stop, so the benefit is partial, and the share of missed defects is almost unchanged (6.5% against 7.2% at 300 operations). The first version could only conjecture this benefit.

5.12 Coverage of neighboring valves

A wrong-valve action that moves a neighbor without a device is invisible to the layer unless the neighbor is covered. With the NFC tap-before-touch check the layer already cuts the collateral events from 10.1 to 1.4 per 1000 transfers, and without any device on the neighbors the undetected events are 0.83 per 1000 (existing checks alone: 5.07). The undetected events fall in proportion to the coverage (Fig. 10c) and almost vanish when every neighbor carries a device (0.01). Coverage of neighbors is therefore a design choice with a quantifiable return, and a partial coverage, for example of the valves adjacent to the route, removes a proportional part.

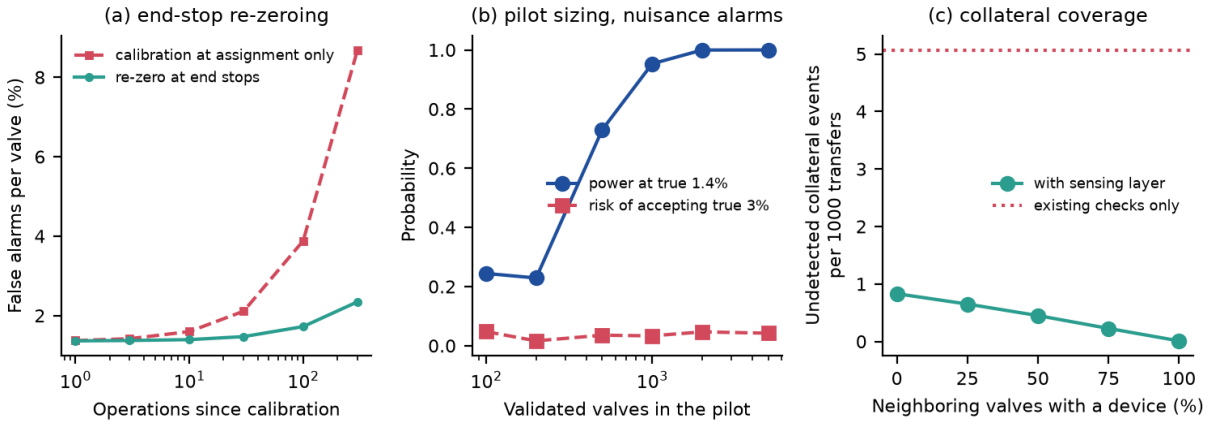


Fig. 10. (a) False alarms against operations since calibration with and without end-stop re-zeroing. (b) Pilot sizing for nuisance alarms: power at the modeled rate and risk of accepting an

unacceptable rate. (c) Undetected collateral events against the fraction of neighboring valves that carry a device.

5.13 Pilot sizing and acceptance criteria

Table 5 gives exact binomial acceptance numbers for the nuisance-alarm rate. A pilot of 1000 valves with an acceptance number of 20 has a power of 95% at the modeled 1.4% and a risk of 3.3% of accepting a device whose true rate is 3%; smaller pilots cannot separate the two (100 to 200 valves have a power below 23%), and the extra information of more than 2000 valves is small. A pilot on one route therefore needs about 1000 valve validations, which at 14 valves per validation is about 70 transfers, or the same valves validated repeatedly across several routes.

Valves validated N	Acceptance number	Power at a true rate of 1.4%	Risk of accepting a true rate of 3%	Upper 95% limit if 1.4% is observed
100	≤ 0	24%	4.8%	5.4%
200	≤ 1	23%	1.6%	4.3%
500	≤ 8	73%	3.5%	2.9%
1,000	≤ 20	95%	3.3%	2.3%
2,000	≤ 47	100%	4.7%	2.0%
5,000	≤ 129	100%	4.2%	1.8%

Table 5. Pilot acceptance test for the nuisance-alarm rate: accept if the number of nuisance alarms among the N validated valves is at most the acceptance number (consumer risk 5% at a true rate of 3%).

For missed defects, the injected-defect test of Table 6 needs about 80 deliberately misplaced valves to demonstrate a miss rate below 15% with a power of 80% at the modeled 6% (150 valves give 98%). The accuracy of the device is shown by repeated readings of a valve held at a known position: with 100 readings the 95% interval for the standard deviation is 0.88 to 1.16 times the observed value, with 500 readings 0.94 to 1.07.

Defects injected	Accept if misses	Power at a true miss rate of 6%
20	≤ 0	29%
40	≤ 2	57%
80	≤ 6	80%
150	≤ 15	98%

Defects injected	Accept if misses	Power at a true miss rate of 6%
300	≤ 34	100%

Table 6. Pilot acceptance test for missed defects: injected defects detected or missed (consumer risk 5% at a true miss rate of 15%).

6. Discussion

Four findings are useful for specification. First, the headline reduction should be quoted as a distribution, a factor of about 10 (plausible range 4 to 28) rather than the nominal 14, and as conditional on operators acting on the alarms: at a 90% response the factor is 5.6. The benefit is therefore a property of the whole layer, sensor plus people plus procedure, and a pilot must measure the response, not only the sensor. Second, an 8% alarm tolerance with two-point calibration at every assignment keeps nuisance alarms near 1% per valve with direct sensing and is close to optimal once alarm fatigue is included; end-stop re-zeroing of binary valves is worth specifying because it removes most of the dependence on calibration age; and throttled valves need their own, explicitly derived band. Third, on-site calibration quality, availability and bypass policy, not the radio technology, govern performance: they have the largest correlations with the benefit among the inputs a design controls, so spare devices, rapid swap and a prohibition on unrecorded bypass are part of the barrier. Fourth, the NFC tap-before-touch check addresses a different problem from the sensors (prevention of wrong-valve actions), and coverage of neighboring valves has a proportional return.

The comparison of the two sensing concepts supports sensing at the stem or indicator where this is mechanically feasible. The turn collar remains attractive where only the handwheel is accessible, but the simulation indicates it needs recalibration at nearly every assignment and a wider tolerance.

6.1 Position relative to layer-of-protection practice

A protection layer is credited in layer-of-protection analysis only when it is independent of the initiating event and of other layers, specific, dependable and auditable [9]. The sensing layer is independent of the operator's report by construction, and its time-stamped record is auditable. It is not dependable in the required sense until its availability, accuracy and the response of the people who act on it have been measured; the study shows, in addition, that these are the parameters that decide the benefit. The layer is therefore proposed as an advisory barrier whose value is to be demonstrated by a pilot with the acceptance criteria of Section 5.13, and not as a credited independent protection layer or a safety instrumented function.

6.2 Limitations

The study has important limitations. (i) All input probabilities are assumed priors. The probabilistic sensitivity analysis propagates their uncertainty, but the prior ranges are themselves assumptions, and human-error rates, existing-barrier detection and valve behavior at a real facility may differ by more than the factors used. (ii) Common cause is represented only by a shared multiplier of the operator error rates; common-cause failures of the sensing layer (a radio outage, a mislabeled tag group, one calibration procedure applied to all devices) are not modeled. (iii) The operator-response model is a parametric description with assumed coefficients, not a fitted model, and the optimum tolerance depends on it. (iv) Collateral movement is modeled for the neighbor of a wrong-valve action only, and a covered neighbor is assumed to be detected reliably. (v) Detection of out-of-sequence steps is represented by a single probability rather than a full procedure model. (vi) Security, electromagnetic and mechanical failure modes, hazardous-area certification and the installation effort were not simulated. (vii) The device concept has not been built or tested, and the battery estimate is analytical. (viii) The literature cited for the

human-reliability and incident background was checked for existence and bibliographic data, not read in full for each statement.

6.3 Relation to existing practice and future work

Permanent position transmitters and limit switches provide equivalent information at higher installation cost per valve, and they are the natural choice for valves on critical lines that are operated frequently. The temporary, reassignable layer studied here addresses the complementary case of many manual valves that matter only on some routes or during some operations. The next step is the pilot of Section 5.13, which measures sensor accuracy, nuisance alarms, availability, installation time and operator response on a real route, replaces the assumed inputs with data and allows the model to be calibrated; the probabilistic sensitivity analysis can then be rerun with posterior distributions.

7. Conclusions

An independent sensing layer on existing manual valves reduces the modeled probability of an undetected misalignment or unauthorized movement per transfer by a factor of 14 on a 14-valve route with the nominal assumed inputs (from 4.03% to 0.29%), and by a median factor of 10 (5th to 95th percentile 4 to 28) when the uncertainty of the assumed inputs is propagated. The reduction depends on operators acting on the alarms: it falls to 5.6 at a 90% response. Direct stem or indicator sensing outperforms handwheel turn counting in false alarms and in robustness to calibration age. Alarm tolerance near 8% of stroke, on-site calibration quality with end-stop re-zeroing, spare availability and bypass policy are the main levers, throttled valves need a separate band, and NFC identity checks mainly prevent wrong-valve events. A pilot of about 1000 validated valves with an acceptance number of 20 nuisance alarms and about 80 to 150

injected defects would replace the assumed inputs with data. The layer cannot be credited as a quantified protection layer before that.

Declaration of competing interest

The author has prepared a related commercial proposal in response to a public open-innovation challenge concerning real-time verification of manual valve position and may benefit commercially from the concept described. The author has no other competing interests.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Leon Sandler: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review and editing.

Data availability

The simulation code, the data behind every figure and table, the automated tests and the manuscript sources (version 1.0.0) are archived at https://doi.org/10.5281/zenodo.24046773 (software, all versions) and https://doi.org/10.5281/zenodo.24046771 (manuscript, all versions), and are available at https://github.com/sandlerleon/hydrasense-valve-verification-mc. The study uses no field data.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) to assist with drafting text, writing and testing simulation code and generating figures from that code. After using this

tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

References

[1] A.D. Swain, H.E. Guttmann, Handbook of Human Reliability Analysis with Emphasis on Nuclear Power Plant Applications, NUREG/CR-1278, U.S. Nuclear Regulatory Commission, Washington, DC (1983).

[2] B. Kirwan, A Guide to Practical Human Reliability Assessment, Taylor & Francis, London (1994).

[3] J. Reason, Human Error, Cambridge University Press, Cambridge (1990). https://doi.org/10.1017/cbo9781139062367

[4] B. Kirwan, Human error identification in human reliability assessment. Part 1: Overview of approaches, Applied Ergonomics 23 (1992) 299–318. https://doi.org/10.1016/0003-6870(92)90292-4

[5] J.C. Williams, A data-based method for assessing and reducing human error to improve operational performance, in: Conference Record for 1988 IEEE Fourth Conference on Human Factors and Power Plants, IEEE (1988) 436–450. https://doi.org/10.1109/hfpp.1988.27540

[6] E. Zarei, M. Yazdi, R. Abbassi, F. Khan, A hybrid model for human factor analysis in process accidents: FBN-HFACS, Journal of Loss Prevention in the Process Industries 57 (2019) 142–155. https://doi.org/10.1016/j.jlp.2018.11.015

[7] T. Kletz, P. Amyotte, Case histories and their use in enhancing process safety knowledge, in: What Went Wrong?, Elsevier (2019) 3–17. https://doi.org/10.1016/b978-0-12-810539-9.00001-x

513 [8] S. Mannan (Ed.), Lees' Loss Prevention in the Process Industries: Hazard Identification,
514 Assessment and Control, fourth ed., Butterworth-Heinemann, Oxford (2012); chapter 14,
515 Human factors and human error, pp. 687–751. https://doi.org/10.1016/b978-0-12-397189-
516 0.00014-8
517 [9] Center for Chemical Process Safety, Guidelines for Initiating Events and Independent
518 Protection Layers in Layer of Protection Analysis, Wiley, Hoboken (2015).
519 https://doi.org/10.1002/9781118948743
520 [10] F. Chraim, K. Pister, Wireless valve position monitoring: A MEMS approach, in: IECON
521 2013 – 39th Annual Conference of the IEEE Industrial Electronics Society, IEEE (2013)
522 4016–4021. https://doi.org/10.1109/iecon.2013.6699778
523 [11] Wazoku (InnoCentive), Real-Time Verification of Manual Valve Position during Critical
524 Product Transfers (scouting challenge for YPF),
525 https://ypf.community.innocentive.com/challenge/1ca6a86eaede4555a14d7b1746c9d79e
526 (accessed 7 November 2026).
527 [12] J.C. Laberge, P. Bullemer, M. Tolsma, D.V.C. Reising, Addressing alarm flood situations in
528 the process industries through alarm summary display design and alarm response strategy,
529 International Journal of Industrial Ergonomics 44 (2014) 395–406.
530 https://doi.org/10.1016/j.ergon.2013.11.008
531 [13] IEC 62591, Industrial networks – Wireless communication network and communication
532 profiles – WirelessHART, International Electrotechnical Commission, Geneva (current
533 edition).
534 [14] IEC 61511 series, Functional safety – Safety instrumented systems for the process industry
535 sector, International Electrotechnical Commission, Geneva.

[15] IEC 60079-11, Explosive atmospheres – Part 11: Equipment protection by intrinsic safety "i", International Electrotechnical Commission, Geneva (current edition).

[16] N. Metropolis, S. Ulam, The Monte Carlo method, Journal of the American Statistical Association 44 (1949) 335–341. https://doi.org/10.1080/01621459.1949.10483310

[17] E.B. Wilson, Probable inference, the law of succession, and statistical inference, Journal of the American Statistical Association 22 (1927) 209–212. https://doi.org/10.1080/01621459.1927.10502953

[18] L.D. Brown, T.T. Cai, A. DasGupta, Interval estimation for a binomial proportion, Statistical Science 16 (2001) 101–133. https://doi.org/10.1214/ss/1009213286

[19] L. Sandler, MyUncle: maintenance-lattice simulation framework, GitHub repository, https://github.com/sandlerleon/MyUncle.

[20] S. Marino, I.B. Hogue, C.J. Ray, D.E. Kirschner, A methodology for performing global uncertainty and sensitivity analysis in systems biology, Journal of Theoretical Biology 254 (2008) 178–196. https://doi.org/10.1016/j.jtbi.2008.04.011

[21] IEC 62682, Management of alarm systems for the process industries, International Electrotechnical Commission, Geneva (current edition).